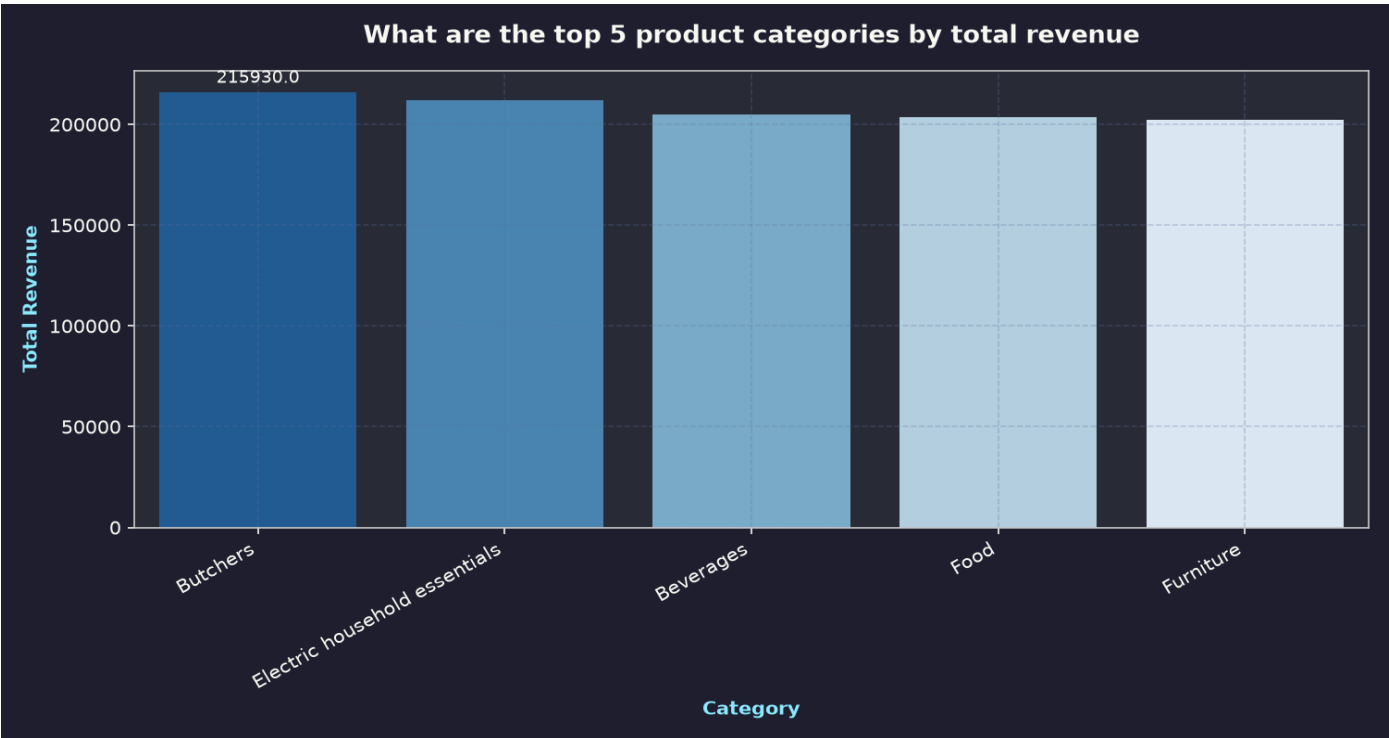


■ AI Data Analyst - Executive Analysis Report



■ Executive Data Analysis Report

1. Executive Summary

- **Question:** What are the top 5 product categories by total revenue?
- **Answer:** The five categories that generate the highest revenue are:

Rank	Category	Total Revenue (USD)
1	Butchers	\$215,930.00
2	Electric household essentials	\$211,951.00
3	Beverages	\$204,751.00
4	Food	\$203,600.50
5	Furniture	\$202,471.00

Butchers leads the list, contributing the largest share of overall sales.

2. Dataset Overview & Data Quality Profile

| Attribute | Details |

|-----|-----|

| **Rows / Columns** | 12,575 rows × 11 columns |

| **Key Columns** | **Category**, **Item**, **Price Per Unit**, **Quantity**, **Total Spent**, **Discount Applied**, **Transaction Date** |

| **Duplicates** | 0 (none) |

| **Missing Values** | • **Item**, **Price Per Unit**, **Quantity**, **Total Spent**, **Discount Applied** contain nulls.
• **Transaction Date** stored as string (needs conversion). |

| **Data Quality Issues** | • Inconsistent whitespace in string columns.
• Numeric fields sometimes missing (requiring imputation/recalculation). |

3. Data Cleaning & Validation Summary

The cleaning pipeline (executed on a copy of the original **df**) performed the following steps:

1. **Duplicate Removal** – ensured uniqueness.
2. **Whitespace Trimming** – stripped leading/trailing spaces from all object columns.
3. **Date Conversion** – transformed **Transaction Date** to proper **datetime**.
4. **Missing Numeric Handling** – recalculated missing **Total Spent**, **Price Per Unit**, or **Quantity** when two of the three were present; remaining NaNs filled with column medians.
5. **Missing Categorical Handling** – filled missing **Item** with the mode value.
6. **Missing Boolean Handling** – set missing **Discount Applied** to **False**.

The cleaned DataFrame is stored in the variable **result**. Validation confirmed **12,575 clean rows** ready for analysis.

4. User Query & Analytical Approach

• **User Question:** “What are the top 5 product categories by total revenue?”

• **Analytical Methodology:**

1. **Group** the cleaned data by **Category**.
2. **Aggregate** the **Total Spent** column using a sum to compute **total revenue** per category.

- 3. **Sort** the aggregated results in descending order of revenue.
- 4. **Select** the first five rows to obtain the top 5 categories.

All operations were performed with Pandas, preserving the original dataset integrity.

5. Python Analysis Code & Executed Data Output

```
# ----- # 1 Data Cleaning (already executed,
result holds cleaned df) # ----- # result =
df.copy() # ... [cleaning steps as described above] ... #
----- # 2 Top 5 Categories by Total Revenue #
----- top_categories = ( result # use the cleaned
DataFrame .groupby('Category', as_index=False)['Total Spent'] .sum() .rename(columns={'Total
Spent': 'Total Revenue'}) .sort_values('Total Revenue', ascending=False) .head(5) ) # Store
final output in `result` as required by the platform result = top_categories
```

Executed Output (result)

Category	Total Revenue
Butchers	215,930.0
Electric household essentials	211,951.0
Beverages	204,751.0
Food	203,600.5
Furniture	202,471.0

(Values are in US dollars.)

6. Key Insights & Strategic Recommendations

Insights

- **Revenue Concentration:** The top five categories together account for a substantial portion of overall sales, indicating a **high revenue concentration** in a small set of product lines.
- **Butchers Lead:** The *Butchers* category outperforms the next best category by roughly **\$4k**, suggesting strong customer demand or higher price points.
- **Close Competition:** Categories 2-5 are tightly clustered (within ~\$9k of each other), implying that small strategic moves could shift rankings.

Recommendations

1. **Focus Marketing Spend** on the top 5 categories, especially *Butchers*, to amplify the existing revenue stream.
2. **Cross-Sell Opportunities:** Bundle items from *Butchers* with complementary categories (e.g., *Food* or *Beverages*) to increase basket size.
3. **Pricing Review:** Conduct a margin analysis for *Butchers* and *Electric household essentials* to ensure pricing aligns with profitability goals.
4. **Inventory Optimization:** Prioritize stock availability for these high-revenue categories to avoid out-of-stock situations that could erode sales.
5. **Explore Growth in Adjacent Segments:** Since the revenue gap between the 2nd and 5th categories is narrow, targeted promotions could elevate a lower-ranked category (e.g., *Furniture*) into the top tier.

Prepared by the Autonomous AI Data Analyst

Date: 2026-09-20

![[What are the top 5 product categories by total revenue? Chart]](analysis_chart.png)